المعهد الوطني للبريد والمواصلات
ⵜⴰⵙⴷⴰⵡⵜ ⵏ ⵜⵙⵍⵉⵜ ⴰ ⵜⵜⵎⵙⵙⴰⵡⴰⵍ
Institut National des Postes et Télécommunications

Interpolation and Polynomial Approximation

Major: M1 Data Science & AI

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Question 1: Calculate $P_1(x)$, and $P_2(x)$, $P_3(x)$, $P_4(x)$.
Solution : let

$$(x_i, f(x_i)) \quad \forall i \in \{0, \ldots, n\}$$
$$\text{we have : } P_n(x_i) = f(x_i) \quad \forall i \in \{0, \ldots, n\}$$
$$P_n(x) = \sum_{k=0}^{n} f(x_k) \cdot L_k(x)$$
$$L_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{x - x_j}{x_k - x_j}, \quad k = 0, \ldots, n$$

for $P_1(x)$:

$$\begin{aligned} P_1(x) &= \sum_{k=0}^{1} f(x_k) L_k(x) \\ &= f(x_0)L_0(x) + f(x_1)L_1(x) \end{aligned}$$

the lagrange basis polynomials are :

$$L_0(x) = \prod_{\substack{j=0 \\ j \neq 0}}^{1} \frac{x - x_j}{x_0 - x_j} = \frac{x - x_1}{x_0 - x_1}$$

$$L_1(x) = \prod_{\substack{j=0 \\ j \neq 1}}^{1} \frac{x - x_j}{x_1 - x_j} = \frac{x - x_0}{x_1 - x_0}$$

Substituting the expressions of $L_0(x)$ and $L_1(x)$ into $P_1(x)$, we get :

$$P_1(x) = f(x_0)\frac{x - x_1}{x_0 - x_1} + f(x_1)\frac{x - x_0}{x_1 - x_0}$$

Pour $P_2(x)$:

$$\begin{aligned} P_2(x) &= \sum_{k=0}^{2} f(x_k) L_k(x) \\ &= f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) \end{aligned}$$

the lagrange basis polynomials are :

$$L_0(x) = \prod_{\substack{j=0 \\ j\neq 0}}^{2} \frac{x - x_j}{x_0 - x_j} = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$L_1(x) = \prod_{\substack{j=0 \\ j\neq 1}}^{2} \frac{x - x_j}{x_1 - x_j} = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$L_2(x) = \prod_{\substack{j=0 \\ j\neq 2}}^{2} \frac{x - x_j}{x_2 - x_j} = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

Substituting the expressions of $L_0(x)$, $L_1(x)$, and $L_2(x)$ into $P_2(x)$, we get :

$$P_2(x) = f(x_0)\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} + f(x_1)\frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} + f(x_2)\frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

For $P_3(x)$:

$$\begin{aligned} P_3(x) &= \sum_{k=0}^{3} f(x_k)L_k(x) \\ &= f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) + f(x_3)L_3(x) \end{aligned}$$

The Lagrange basis polynomials are :

$$L_0(x) = \prod_{\substack{j=0 \\ j\neq 0}}^{3} \frac{x - x_j}{x_0 - x_j} = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}$$

$$L_1(x) = \prod_{\substack{j=0 \\ j\neq 1}}^{3} \frac{x - x_j}{x_1 - x_j} = \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)}$$

$$L_2(x) = \prod_{\substack{j=0 \\ j\neq 2}}^{3} \frac{x - x_j}{x_2 - x_j} = \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)}$$

$$L_3(x) = \prod_{\substack{j=0 \\ j\neq 3}}^{3} \frac{x - x_j}{x_3 - x_j} = \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)}$$

Substituting the expressions of $L_0(x)$, $L_1(x)$, $L_2(x)$, and $L_3(x)$ into $P_3(x)$, we get :

$$P_3(x) = f(x_0)\frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} + f(x_1)\frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}$$
$$+ f(x_2)\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} + f(x_3)\frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}$$

For $P_4(x)$:

$$P_4(x) = \sum_{k=0}^{4} f(x_k)L_k(x)$$
$$= f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) + f(x_3)L_3(x) + f(x_4)L_4(x)$$

The Lagrange basis polynomials are :

$$L_0(x) = \prod_{\substack{j=0 \\ j\neq 0}}^{4} \frac{x-x_j}{x_0-x_j} = \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)}$$

$$L_1(x) = \prod_{\substack{j=0 \\ j\neq 1}}^{4} \frac{x-x_j}{x_1-x_j} = \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)}$$

$$L_2(x) = \prod_{\substack{j=0 \\ j\neq 2}}^{4} \frac{x-x_j}{x_2-x_j} = \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)}$$

$$L_3(x) = \prod_{\substack{j=0 \\ j\neq 3}}^{4} \frac{x-x_j}{x_3-x_j} = \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)}$$

$$L_4(x) = \prod_{\substack{j=0 \\ j\neq 4}}^{4} \frac{x-x_j}{x_4-x_j} = \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)}$$

Substituting the expressions of $L_0(x)$, $L_1(x)$, $L_2(x)$, $L_3(x)$, and $L_4(x)$ into $P_4(x)$, we get :

$$P_4(x) = f(x_0)\frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} + f(x_1)\frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)}$$
$$+ f(x_2)\frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} + f(x_3)\frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)}$$
$$+ f(x_4)\frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)}$$

Question 2: Write a Python Code for this problem and find the Lagrange polynomial for these Points: $(1,3);(10,7);(8,5)$

Solution :

```python
def multiplypoly(poly1,poly2):
    result=[0]*(len(poly1)+len(poly2)-1)
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            result[i+j]+=poly1[i]*poly2[j]
    return result

def computeLk(points,k):
    n=len(points)
    x_points=[p[0] for p in points]

    num=[1.0]
    den=1
    for j in range(n):
        if j!=k:
            num=multiplypoly(num,[-x_points[j],1])
            den*=(x_points[k]-x_points[j])
    return [c/den for c in num]
pts=[(3,0),(4,1)]
print(computeLk(pts,0))
```

[11] ✓ 0.0s Python

```
[4.0, -1.0]
```

```python
def computeP(points):
    n=len(points)
    x_points=[p[0] for p in points]
    y_points=[p[1] for p in points]
    p=[0.0]*n
    for k in range(n):
        Lk=computeLk(points,k)
        for i in range(len(Lk)):
            p[i]+=y_points[k]*Lk[i]

    print(f"p_{n-1}(x)=", end="")
    style=" ¹²³⁴⁵⁶⁷⁸⁹"
    for d in range(len(p)-1,-1,-1):
        if round(p[d], 2) == 0:
            continue
        if (d==0) :
            print(f"{p[d]:.2f}",end=" ")
        else:
            print(f"{p[d]:.2f}x{style[d]}",end=" + ")

pts=[(1,3),(10,7)]
computeP(pts)
```

[13] ✓ 0.0s Python

```
p_1(x)=0.44x¹ + 2.56
```

```python
def all_Lk(points):
    n=len(points)

    for k in range(n):
        Lk=computeLk(points,k)

        print(f"L_{k}(x)=", end="")
        style=" ¹²³⁴⁵⁶⁷⁸⁹"
        for d in range(len(Lk)-1,-1,-1):
            if round(Lk[d], 2) == 0:
                continue
            if (d==0) :
                print(f"{Lk[d]:.2f}",end=" ")
            else:
                print(f"{Lk[d]:.2f}x{style[d]}",end=" + ")
        print()


pt=[(1,3),(10,7),(8,5)]
all_Lk(pt)
```

[28] ✓ 0.0s Python

```
L_0(x)=0.02x² + -0.29x¹ + 1.27
L_1(x)=0.06x² + -0.50x¹ + 0.44
L_2(x)=-0.07x² + 0.79x¹ + -0.71
```

```python
def lagrange_interpolation(points):
    all_Lk(points)
    print()
    computeP(points)
pt=[(1,3),(10,7),(8,5)]
lagrange_interpolation(pt)
```

[29] ✓ 0.0s Python

```
L_0(x)=0.02x² + -0.29x¹ + 1.27
L_1(x)=0.06x² + -0.50x¹ + 0.44
L_2(x)=-0.07x² + 0.79x¹ + -0.71

p_2(x)=0.08x² + -0.43x¹ + 3.35
```